

Craigslist Clone UML Diagram (For Sale Ads + Search & Messaging)

Extended UML including posting, searching, filtering, and user messaging for the **For Sale** section.

1. Use Case Diagram

```
%% UML Use Case Diagram with search and messaging
flowchart TD
    User([User])
    Admin([Admin])
    System((Craigslist Clone))

    User -->|create| UC1[Post For Sale Ad]
    User -->|update| UC2[Edit Ad]
    User -->|delete| UC3[Delete Ad]
    User -->|view| UC4[View & Search Ads]
    User -->|filter| UC5[Filter Ads]
    User -->|send| UC6[Message Seller]
    User -->|read| UC7[View Inbox]
    Admin -->|moderate| UC8[Moderate Ads & Messages]

    UC1 --> System
    UC2 --> System
    UC3 --> System
    UC4 --> System
    UC5 --> System
    UC6 --> System
    UC7 --> System
    UC8 --> System
```

2. Class Diagram

```
classDiagram
    class User {
        +Long id
        +String name
        +String email
    }
```

```

        +String passwordHash
        +Date createdAt
        +List<Ad> ads
        +List<Message> messages
    }

    class Ad {
        +Long id
        +String title
        +String description
        +Double price
        +String condition
        +String location
        +List<Image> images
        +Category category
        +Date createdAt
        +Date updatedAt
        +AdStatus status
    }

    class Category {
        +Long id
        +String name
        +String slug
    }

    class Image {
        +Long id
        +String url
        +String thumbnailUrl
    }

    class Message {
        +Long id
        +User sender
        +User receiver
        +Ad ad
        +String body
        +Boolean isRead
        +Date sentAt
    }

    class SearchFilter {
        +String keyword
        +String category
        +Double minPrice
        +Double maxPrice
        +String location
    }

```

```

        +String sortBy
    }

    class AdStatus {
        <<enumeration>>
        ACTIVE
        PENDING
        SOLD
        REMOVED
    }

    User --> Ad : owns >
    Ad --> Category : belongs to >
    Ad --> Image : has many >
    User --> Message : sends/receives >
    Ad --> Message : related to >

```

3. Sequence Diagram – Searching & Messaging

a) Searching / Filtering Ads

```

sequenceDiagram
    actor User
    participant AngularApp as Angular Frontend
    participant API as Spring Boot API
    participant ES as Elasticsearch

    User ->> AngularApp: Enter search keywords / filters
    AngularApp ->> API: GET /api/v1/listings?
    query=laptop&minPrice=500&maxPrice=2000
    API ->> ES: Perform full-text + filter query
    ES -->> API: Return matched ads
    API -->> AngularApp: JSON list of ads
    AngularApp -->> User: Display search results

```

b) Messaging Seller

```

sequenceDiagram
    actor Buyer
    participant AngularApp as Angular Frontend
    participant API as Spring Boot API
    participant DB as PostgreSQL
    participant Seller

```

```
Buyer ->> AngularApp: Write message to seller
AngularApp ->> API: POST /api/v1/messages {adId, body}
API ->> DB: Save message (senderId, receiverId, adId, body)
API -->> AngularApp: 201 Created (message saved)
Seller ->> AngularApp: Fetch messages
AngularApp ->> API: GET /api/v1/messages/inbox
API ->> DB: Retrieve messages for seller
API -->> AngularApp: JSON message list
AngularApp -->> Seller: Display inbox
```

4. New Entities Added

- **Message:** facilitates buyer-seller communication.
- **SearchFilter:** defines parameters for search queries.

5. Updated API Endpoints (new)

```
GET /api/v1/listings?query=&category=&minPrice=&maxPrice=&location=&sortBy=
POST /api/v1/messages
GET /api/v1/messages/inbox
GET /api/v1/messages/sent
```

6. Next Steps

- Add UML for **Admin Moderation Workflow**.
- Include **Notification System** (for new messages and ad status updates).
- Extend to **Saved/Favorite Ads** feature.

(Would you like me to add the Admin Moderation UML next?)